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Sugar: A Python framework for bioinformatics

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Software

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Summary

Modern bioinformatics requires the use of a variety of tools (e.g. BLAST for homology detection, Altschul et al., 1990) and databases (e.g. GenBank, Benson et al., 2012). Custom scripts often need to use the results of these tools, which are available in different formats.

Sugar is a Python framework for bioinformatics and aims to facilitate rapid application development. The package allows to read and write various sequence and annotation formats, i.e. FASTA, GenBank, Stockholm, GFF, GTF, BLAST and others. Since sugar uses a plugin system for reading and writing, new file formats can be added not only within the sugar package, but also within other packages, allowing for low barrier inclusion of new formats. Sugar includes classes for representing DNA/RNA sequences and annotations. The main functionality is exposed through methods of these classes and is therefore readily available.

Sugar can be used by researchers and students of bioinformatics alike. The package has already been used as a library in the AnchoRNA package (Eulenfeld et al., 2025) and in several scripts that form the basis of the Ritsch et al. (2024) publication. Sugar can be installed from PyPI. Online documentation and tutorials are available on the GitHub project site.

Statement of need

Other well-known frameworks for bioinformatics are Biopython (Cock et al., 2009) and Biotite (Kunzmann & Hamacher, 2018). Launched in 2000, Biopython contains a large collection of freely available tools. In contrast, sugar tries to focus on the basics of IO as well as sequence and annotation (resp. feature) manipulation. In Table 1, we compare the code for reading a FASTA sequence file using sugar and Biopython, respectively. In addition, the sugar code example demonstrates a typical bioinformatics file manipulation task, which is not easily possible with Biopython alone due to its lack of a GFF writer. The excellent Biotite package also handles DNA/RNA sequences and has an additional focus on protein structures. Sugar should not be seen as an alternative to these other packages, but rather as a complement. Therefore, adapters for these packages are provided, which can easily convert sugar sequence objects into the corresponding objects of the Biopython and Biotite packages, and vice versa. In addition, sugar annotation objects can be converted to corresponding objects of the DNA features viewer package (Zulkower & Rosser, 2020) for easy plotting.

Table 1: Comparison of code to read a FASTA file using sugar (left) and Biopython (right). The code example using sugar also demonstrates reading of a BLAST result file, attaching the hit features to the sequences while discarding features belonging to different sequences, and writing the sequences with the corresponding features to a GFF file including a FASTA directive.

`from sugar import read, read_fts`
`seqs = read('seqs.fasta')`
`fts = read_fts('hits.blastn')`
`seqs.fts = fts`
	`from Bio import SeqIO`
	`seqs = list(SeqIO.parse('seqs.fasta', 'fasta'))`

```
seqs.write('seqs_annotated.gff')
```


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